

CRICGEEK

Expression Quality Score (EQS)

Technical Design, Scoring Methodology & Implementation To-Do

Internal technical specification for the planned CricGeek content-quality system, Voice-to-Commentary workstream, and database-design workstream.

Document	EQS Technical Design & To-Do
Product	CricGeek Network LLP
Status	Technical planning / implementation specification
Version	1.0

1. What EQS Is

EQS (Expression Quality Score) is CricGeek's planned system for evaluating the quality, credibility, expression, reasoning, originality, and suitability of cricket content. It is not simply an AI detector, sentiment score, or grammar checker.

The system combines specialist checks, structured AI outputs, cricket-stat verification, plagiarism checks, and Writer DNA context. The final EQS is produced only after the component results are available.

Core principles

- No single signal should decide whether an article is good or bad.
- Negative sentiment is not automatically poor quality; strong criticism can be high quality when it is reasoned and constructive.
- Positive wording is not automatically high quality; sarcasm, vague praise, unsupported claims, or manipulation must still be considered.
- AI-generated/AI-assisted probability is a signal, not an automatic rejection rule.
- Cricket statistics must be verified separately using trusted cricket data/API sources.
- Internal plagiarism and external plagiarism are separate systems.

2. Planned EQS Processing Architecture

Stage	Purpose	Mechanism
1	Article input and metadata	CricGeek application/backend
2	Writer DNA	Fan / Analyst / Debater / Storyteller profile
3	Fast checks	Rules and deterministic checks
4	Language analysis	Gemini 3.5 Flash or approved successor
5	Claim extraction	Structured AI output
6	Cricket verification	Dedicated cricket data/API + verification logic
7	Internal plagiarism	Exact match + embeddings/vector similarity
8	External plagiarism	Dedicated plagiarism API/service
9	Component results	Normalize and store individual results
10	Final EQS	DNA-aware interpretation + deterministic guardrails
11	Output	Final score + component breakdown + confidence + flags

Key rule: the final model receives the structured results from the specialist modules. It should not independently invent the cricket-stat or plagiarism findings.

3. AI Model Role

The current planned primary language-analysis model is **Gemini 3.5 Flash** (model ID **gemini-3.5-flash**). Google's current documentation lists it as a stable model with structured outputs and tool/function support. The production model should remain configurable so CricGeek can benchmark newer or lower-cost models later.

Use a smaller/cost-efficient model for high-volume sub-tasks only if CricGeek's own benchmark shows acceptable quality.

4. Writer DNA

EQS must be Writer-DNA-aware because different cricket writers have different purposes. The scoring methodology should support mixed profiles as well.

DNA	Typical purpose	Higher-priority considerations
Fan	Passion, perspective, emotional connection	Originality, expression, coherence, relevance, constructive tone, authenticity
Analyst	Data, tactics, technical interpretation	Stat accuracy, evidence, reasoning, relevance, clarity
Debater	Arguments, comparisons, viewpoints	Reasoning, evidence, constructive disagreement, coherence, relevance
Storyteller	Narrative, context, human angle	Originality, narrative flow, coherence, expression, relevance, engagement

5. EQS Evaluation Dimensions

Dimension	What it evaluates
Originality	Whether the article adds its own information, interpretation, framing, or expression rather than merely reproducing sources.
Expression quality	Clarity, readability, precision, vocabulary, sentence quality, and communication of the intended point.
Coherence	Logical flow, paragraph relationships, beginning/middle/end where applicable, and absence of confusing jumps.
Relevance & focus	Connection to the topic; avoidance of filler, repetition, and vague statements.
Sentiment	Emotional direction in context; never a simple positive=good / negative=bad rule.
Sarcasm & intent	Contextual understanding of sarcasm, mockery, rhetorical language, and literal-vs-intended meaning.
Toxicity / personal attacks	Insults, harassment, abusive language, targeted attacks, or unnecessarily hostile expression.
Constructive criticism	Whether criticism explains a weakness or disagreement rather than merely attacking a person.
Reasoning	Quality of arguments, cause/effect logic, comparisons, inference, and explanation.
Evidence quality	Whether important claims are supported by statistics, examples, references, or clear reasoning.
Statistical claims	Claims requiring separate cricket-stat verification.
Writer-DNA suitability	Whether the article succeeds according to its intended writer type.
AI assistance signal	Likelihood/indicators of AI-generated or AI-assisted content; treated only as a signal.

6. Specialist Systems That Must Be Separate

System	Required role
Cricket Statistics Verification	Extract factual/statistical claims and verify them against trusted cricket data/API sources and CricGeek's verified data.
Internal Plagiarism	Compare against CricGeek's own content corpus using exact matching and/or embeddings/vector similarity.
External Plagiarism	Use a dedicated external plagiarism provider/API and store/interpret its evidence.

7. Structured Intermediate Results

Each module should return structured data rather than only prose. A practical common structure is:

Field	Purpose
component	Module name
score	Normalized score, typically 0–100 where applicable
confidence	Confidence in result

Field	Purpose
evidence	Supporting evidence
flags	Issues requiring attention
claims	Extracted claims/verification state when applicable
source	Provider/module
version	Model/rule/API version

8. Final EQS Scoring Methodology

The final EQS is **Writer-DNA-aware**. Individual component results are normalized first. The scoring engine then applies the appropriate weights for the writer's DNA and article context. Gemini 3.5 Flash can interpret the structured evidence at the final layer, while deterministic rules protect critical constraints and reproducibility.

Conceptual flow: Article → Writer DNA → specialist analysis → individual results → normalization → DNA/context weighting → final Gemini interpretation → deterministic guardrails → Final EQS + confidence + explanation + flags.

Exact numerical weights and thresholds should be calibrated using a human/editorial benchmark rather than permanently hard-coded at this planning stage.

9. Benchmark & Calibration

- Build a benchmark set of roughly 200–500 cricket articles/examples.
- Include Fan, Analyst, Debater, Storyteller, and mixed-DNA cases.
- Include sarcasm, strong criticism, toxicity, incorrect statistics, AI paraphrasing, copied text, vague writing, high-quality human writing, and mixed-quality articles.
- Create human/editorial ground truth for component quality and overall quality.
- Compare candidate models and scoring configurations against that ground truth.
- Measure false positives/negatives, especially for sarcasm, toxicity, originality, plagiarism, and stat verification.
- Version the model, rules, thresholds, and scoring configuration for auditability.

10. Voice-to-Commentary — To-Do Design

Voice-to-Commentary is a separate CricGeek system that converts a creator's spoken cricket commentary into usable/live commentary while preserving meaning and style.

Stage	Planned function
1. Microphone	Capture creator speech.
2. Streaming STT	Convert speech to text with low latency.
3. Cleanup	Correct obvious transcription issues without changing intended meaning.
4. Commentary transformation	Use a low-latency language model to turn the transcript into concise cricket commentary while preserving style.
5. Output	Send the commentary to the CricGeek live commentary/content API.
6. Target	Design toward approximately 15–30 seconds or faster end-to-end, subject to testing.

11. Database Design — To-Do Design

The database must support content, writers, Writer DNA, EQS results, verification evidence, plagiarism results, model versions, commentary data, and auditability. The exact schema should be finalized after reviewing the current CricGeek application.

Area	Required support
Content	Articles, drafts, published versions, metadata, author relationships.
Writer DNA	Writer profile, DNA category/weights, classification history/versioning.
EQS	Final score, component scores, confidence, flags, explanation, scoring version.
Verification	Claims, status, source/provider, evidence, timestamps.
Plagiarism	Internal similarity results and external API results.
Models	Model/provider/version used for each analysis.
Voice	Transcript, transformed commentary, status/timestamps where required.
Audit	Processing events, failures, retries, changes, traceability.
Performance	Indexes, caching, query patterns, backups, security and scaling.

12. Implementation To-Do

Priority	Workstream	Main deliverables
1	EQS	Modular EQS pipeline; Writer DNA; structured component outputs; cricket-stat verification; internal pl
2	Voice-to-Commentary	Microphone → streaming STT → cleanup → commentary transformation → CricGeek API; latency, a
3	Database Design	Review current application; design schema, relationships, indexes, audit/versioning, query patterns, l

13. Engineering Principles

- Keep specialist systems modular so components can be replaced or improved independently.
- Use structured JSON between modules.
- Keep model/provider configuration outside core business logic where practical.
- Cache reusable results to reduce API calls and operating cost.
- Do not treat an LLM as the authoritative source for cricket statistics.
- Store evidence and version information so an EQS can be explained later.
- Route low-confidence or materially conflicting results to human review where appropriate.
- Benchmark before changing production models, weights, or thresholds.

This document captures the planned architecture discussed for CricGeek. Exact API contracts, numerical weights, thresholds, database schema, providers, and production model versions should be finalized during implementation and testing.